

Convex Optimization: Classroom Exercises

Classroom Exercise 1

Primal Problem:

$$\min_x \frac{1}{2} \|x\|^2 \quad \text{subject to} \quad Ax = b$$

1. Form the Lagrangian

We introduce a vector of Lagrange multipliers ν for the equality constraints:

$$L(x, \nu) = \frac{1}{2} \|x\|^2 + \nu^T (Ax - b)$$

2. Compute Stationarity Condition

To find the optimal x , we take the gradient of the Lagrangian with respect to x and set it to zero:

$$\nabla_x L(x, \nu) = x + A^T \nu = 0 \implies x^* = -A^T \nu$$

3. Derive the Dual Function

The dual function $g(\nu)$ is found by substituting the stationary point x^* back into the Lagrangian:

$$\begin{aligned} g(\nu) &= \frac{1}{2} \| -A^T \nu \|^2 + \nu^T (A(-A^T \nu) - b) \\ &= \frac{1}{2} \nu^T A A^T \nu - \nu^T A A^T \nu - \nu^T b \\ &= -\frac{1}{2} \nu^T A A^T \nu - \nu^T b \end{aligned}$$

4. Write the Dual Problem

The dual problem is the maximization of the dual function:

$$\max_{\nu} \left(-\frac{1}{2} \nu^T A A^T \nu - \nu^T b \right)$$

Classroom Exercise 2

Primal Problem:

$$\min_x \sum_i x_i \log x_i \quad \text{subject to} \quad \sum_i x_i = 1$$

1. Write the Lagrangian

We introduce a scalar multiplier λ for the equality constraint:

$$L(x, \lambda) = \sum_i x_i \log x_i + \lambda \left(\sum_i x_i - 1 \right)$$

2. Derive Stationary Condition

Taking the partial derivative with respect to each component x_i and setting it to zero:

$$\begin{aligned} \frac{\partial L}{\partial x_i} &= \log x_i + 1 + \lambda = 0 \\ \log x_i &= -1 - \lambda \end{aligned}$$

3. Compute Optimal x_i

Solving for x_i :

$$x_i^* = e^{-1-\lambda}$$

Since all x_i are equal and must sum to 1 ($\sum_{i=1}^n x_i = 1$), the solution is the uniform distribution:

$$x_i^* = \frac{1}{n}$$